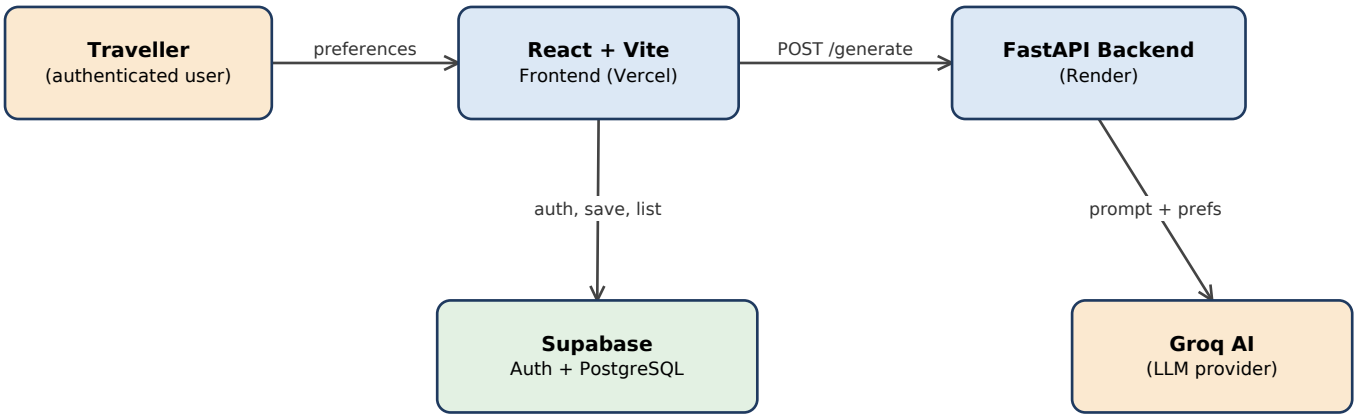
# Data Flow Diagram (DFD)

AI/ML and Generative AI Track • Project Documentation

| | | | |
|---|---|---|---|
| **Project** | Trip Genius AI (VoyagePlan Travel Planner) | **Date** | 26 October 2026 |
| **Submitted by** | Kumkum Thakur (individual project) | **Hall Ticket No.** | 0157CS241131 |
| **College** | LNCTS | **University** | RGPV |

## Level 0 — context diagram




*Responses travel back along the same paths: Groq returns generated itinerary text to FastAPI, FastAPI returns the response schema to the frontend, and the frontend renders the itinerary and the My Trips list to the traveller.*

## Data flow table

| From | Process / destination | Data passed |
|---|---|---|
| Traveller to React/Vite frontend | Planner form | Destination, dates, interests, budget, pace |
| Frontend to FastAPI | POST /api/itineraries/generate | Validated trip preferences (JSON) |
| FastAPI to Groq | Generate itinerary | Prompt / context derived from preferences (server-side API key) |
| Groq to FastAPI | Generated response | Itinerary content |
| FastAPI to frontend | Itinerary response | Days, activities, preferences, status / notice |
| Frontend and Supabase | Authentication and saved trips | User session / token; trip data tied to the user |
| Frontend to traveller | Itinerary and My Trips views | Rendered plan and saved journey list |

## Data stores and external entities

- **External entities:** traveller / authenticated user; Groq AI provider.
- **Data stores:** Supabase Auth (identities, sessions) and Supabase PostgreSQL (saved trips).
- **Security note:** provider API keys stay on the backend (Render environment variables); the browser only holds the public Supabase anon key.